

Configure ipv6 addressing

1. Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).

Static IPv6 Address Configuration

I assigned a static IPv6 address to my **Wi-Fi adapter** using Command Prompt.

Command used:

```
netsh interface ipv6 set address "Wi-Fi" 2001:db8:1::10
```

I then verified the configuration using:

```
ipconfig
```

The IPv6 address was displayed under the **Wireless LAN adapter Wi-Fi**, confirming the configuration.

Static IPv6 Address: 2001:db8:1::10

Network Adapter: Wi-Fi

Verification Command: ipconfig

2. Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address.

IPv6 Unique Local Address Configuration

I configured a Unique Local IPv6 address on my system using the fd prefix. I assigned the address fd12:3456:789a::1 to my computer's Wi-Fi adapter. I then configured another device on the same LAN with the IPv6 address fd12:3456:789a::2. Both devices were connected to the same local network. Finally, I used the ping command to test connectivity between the two devices. The successful ping replies confirmed that communication was working using the Unique Local IPv6 addresses.

3. Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect.

Temporary IPv6 Address Configuration

I configured a temporary IPv6 address on my system and checked the IPv6 configuration using the ipconfig command. I recorded the IPv6 address before disconnecting and reconnecting the network. After reconnecting to the network, I ran ipconfig again and observed that the temporary IPv6 address had changed. I also checked the address after restarting the system and found that a different temporary IPv6 address was assigned. This shows that temporary IPv6 addresses can change after a network reconnection or system reboot, which helps provide better privacy by preventing the same temporary address from being used continuously.

4. Simulate a Zomato-style search: Write a small script (any language) that stores an array of restaurant names and their IPv6 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv6 address.

Zomato-Style Restaurant Search Using JavaScript

```
// Store restaurant names and their IPv6 addresses
const restaurants = [
  { name: "Dominos", ipv6: "2001:db8:1::10" },
  { name: "Pizza Hut", ipv6: "2001:db8:1::20" },
  { name: "McDonalds", ipv6: "2001:db8:1::30" },
  { name: "Subway", ipv6: "2001:db8:1::40" },
  { name: "Burger King", ipv6: "2001:db8:1::50" }
];

// Get search input from user
const search = prompt("Enter restaurant name:").toLowerCase();

// Find matching restaurant
const result = restaurants.filter(restaurant =>
  restaurant.name.toLowerCase().includes(search)
);

// Display result
if (result.length > 0) {
  result.forEach(restaurant => {
    console.log("Restaurant:", restaurant.name);
    console.log("IPv6 Address:", restaurant.ipv6);
  });
} else {
  console.log("Restaurant not found.");
}
```

Sample Output

Enter restaurant name: pizza

Restaurant: Pizza Hut

IPv6 Address: 2001:db8:1::20

Description: This JavaScript program stores restaurant names with their IPv6 addresses. It accepts a partial restaurant name from the user and searches the list using `includes()`. If a match is found, the restaurant name and corresponding IPv6 address are displayed.

5. Use ChatGPT or Copilot to generate a sample IPv6 subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv6 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.

IPv6 Subnetting Scenario

AI-Generated Scenario:

Suppose a company has been assigned the IPv6 network:

2001:db8:1234::/48

The network needs to be divided into at least 3 subnets.

Subnet 1 – Administration:

2001:db8:1234:1::/64

Subnet 2 – Development:

2001:db8:1234:2::/64

Subnet 3 – Sales:

2001:db8:1234:3::/64

I selected Subnet 1 for my system.

IPv6 address assigned to my system:

2001:db8:1234:1::10

Prefix Length:

64

Configuration Steps in Windows:

1. Open Command Prompt as Administrator.
2. First, identify the network interface:
netsh interface ipv6 show interfaces
3. Add the IPv6 address to the Ethernet interface:
netsh interface ipv6 add address "Ethernet" 2001:db8:1234:1::10/64
4. Verify the IPv6 address:
ipconfig
5. The assigned IPv6 address should appear under the Ethernet adapter.

6. Test the IPv6 configuration using:
ping ::1

Result:

The IPv6 address 2001:db8:1234:1::10 was manually assigned to my system from the Administration subnet 2001:db8:1234:1::/64.